

Case Studies

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[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files

```
angela_jose@LAPTOP-B3CJO92Q:~$ mkdir linux_task1
angela_jose@LAPTOP-B3CJO92Q:~$ cd linux_task1
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 angela_jose angela_jose 0 Aug  7 03:58 file1.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug  7 03:58 file2.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug  7 03:58 file3.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug  7 03:58 file4.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug  7 03:58 file5.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ nano file1.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ cat file1.txt
Hello World

angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ rm file5.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories

```
angela_jose@LAPTOP-B3CJO92Q:~$ mkdir linux_task2
mkdir: linux_task2: File exists
angela_jose@LAPTOP-B3CJO92Q:~$ cd linux_task2
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ touch source.txt data.txt notes.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ nano source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ mkdir backup
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ cp source.txt backup
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls backup
source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ cp data.txt backup/data_backup.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls backup
data_backup.txt source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ mv notes.txt backup/
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls
backup data.txt source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls backup
data_backup.txt notes.txt source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ mv source.txt original.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls
backup data.txt original.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ ls backup
data_backup.txt notes.txt source.txt
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files
angela_jose@LAPTOP-B3CJO92Q:~/linux_task2$ |
```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file.

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_task3
mkdir: linux_task3: File exists
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_task3
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ touch students.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ nano students.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ cat students.txt
Angela
Smija
Joel
Riya
Advika
Andrea
Tianne
Evelyn
Ann
Akhil
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ head -5 students.txt
Angela
Smija
Joel
Riya
Advika
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ head students.txt
Angela
Smija
Joel
Riya
Advika
Andrea
Tianne
Evelyn
Ann
Akhil
```

```
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ head -5 students.txt
Angela
Smija
Joel
Riya
Advika
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ head students.txt
Angela
Smija
Joel
Riya
Advika
Andrea
Tianne
Evelyn
Ann
Akhil
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ tail -5 students.txt
Andrea
Tianne
Evelyn
Ann
Akhil
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ tail students.txt
Angela
Smija
Joel
Riya
Advika
Andrea
Tianne
Evelyn
Ann
Akhil
angela_jose@LAPTOP-B3CJ092Q:~/linux_task3$ |
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir organization
angela_jose@LAPTOP-B3CJ092Q:~$ cd organization
angela_jose@LAPTOP-B3CJ092Q:~/organization$ touch file1.txt file2.txt file3.txt imagge1.jpg image2.jpg
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls
file1.txt file2.txt file3.txt image2.jpg imagge1.jpg
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mkdir documents
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mkdir images
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mv *.txt documents/
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls documents
file1.txt file2.txt file3.txt
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mv *.jpg images/
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls images
image2.jpg imagge1.jpg
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls
documents images
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mv documents/file1.txt documents/notes.txt
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls documents
file2.txt file3.txt notes.txt
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mv images/imagel.jpg images/photo.jpg
mv: cannot stat 'images/imagel.jpg': No such file or directory
angela_jose@LAPTOP-B3CJ092Q:~/organization$ mv images/imagge1.jpg images/photo.jpg
angela_jose@LAPTOP-B3CJ092Q:~/organization$ ls images
image2.jpg photo.jpg
angela_jose@LAPTOP-B3CJ092Q:~/organization$ tress
Command 'tress' not found, did you mean:
  command 'tess' from snap tess (0.6.3)
  command 'stress' from deb stress (1.0.7-2)
See 'snap info <snapname>' for additional versions.
angela_jose@LAPTOP-B3CJ092Q:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/organization$ |
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir Project
mkdir: Project: File exists
angela_jose@LAPTOP-B3CJ092Q:~$ cd Project
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir source/
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir source/c/
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir source/python/
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir docs/
mkdir: docs/: File exists
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir backup/
mkdir: backup/: File exists
angela_jose@LAPTOP-B3CJ092Q:~/Project$ touch source/c/file1.c
angela_jose@LAPTOP-B3CJ092Q:~/Project$ touch source/c/file2.c
angela_jose@LAPTOP-B3CJ092Q:~/Project$ touch source/c/file3.c
angela_jose@LAPTOP-B3CJ092Q:~/Project$ ls source/c/
file1.c file2.c file3.c
angela_jose@LAPTOP-B3CJ092Q:~/Project$ touch source/python/program1.py source/python/program2.py
angela_jose@LAPTOP-B3CJ092Q:~/Project$ ls source/python/
program1.py program2.py
angela_jose@LAPTOP-B3CJ092Q:~/Project$ tree
.
├── Project
│   ├── sources
│   ├── backup
│   ├── docs
│   └── source
│       ├── c
│       │   ├── file1.c
│       │   ├── file2.c
│       │   └── file3.c
│       └── python
│           ├── program1.py
│           └── program2.py
└── sources
    ├── c
    └── python
```



```

└─ python

11 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir backup
mkdir: backup: File exists
angela_jose@LAPTOP-B3CJ092Q:~/Project$ ls
Project backup docs source sources
angela_jose@LAPTOP-B3CJ092Q:~/Project$ mkdir backup
mkdir: backup: File exists
angela_jose@LAPTOP-B3CJ092Q:~/Project$ touch backup/data.txt
angela_jose@LAPTOP-B3CJ092Q:~/Project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
angela_jose@LAPTOP-B3CJ092Q:~/Project$ rm backup/data.txt
angela_jose@LAPTOP-B3CJ092Q:~/Project$ ls backup
angela_jose@LAPTOP-B3CJ092Q:~/Project$ rmdir backup
angela_jose@LAPTOP-B3CJ092Q:~/Project$ ls
Project docs source sources
angela_jose@LAPTOP-B3CJ092Q:~/Project$ tree
.
├── Project
│   ├── sources
│   └── python
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py
└── sources
    ├── c
    └── python

10 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/Project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions

```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_task4
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_task4
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ touch public.txt private.txt script.s
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano public.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano private.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod 600 private.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod 644 public.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod +x script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ./script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ |

```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories


```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir permission_lab
angela_jose@LAPTOP-B3CJ092Q:~$ cd permission_lab
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ touch test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ nano test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u-r test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
--w-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u+r test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u-w test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ nano test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u+w text.txt
chmod: cannot access 'text.txt': No such file or directory
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ |

```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp

```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_task6
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_task6
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ touch important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ nano important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mkdir backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp important.txt backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important.txt
└── important.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ rm important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 1 file
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt
cp: missing destination file operand after 'backup/important_backup.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt .
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup  important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

```

```

├── important_backup.txt

2 directories, 1 file
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt
cp: missing destination file operand after 'backup/important_backup.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt .
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup  important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mv important_backup.txt important_restored.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cat important_restored.txt
Welcome to Linux
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_project
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_project
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ mkdir source backup docs
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ touch source/main.c source/helper.c source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls source/
README.txt  helper.c  main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ nano source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ nano source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Mangement
Directory Management
Permissions
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tail -5 source/README.txt
Docs Folder
Learning Commands
Operating Systems
End of File

angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/main.c backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup/
main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
backup  docs  source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt
cp: missing destination file operand after 'source/README.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt docs/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls docs
README.txt
```



```

angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/main.c backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup/
main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
backup docs source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt
cp: missing destination file operand after 'source/README.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt docs/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls docs
README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ mv source/helper.c source/functions.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls source
README.txt functions.c main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ chmod 644 source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 angela_jose angela_jose 159 Aug 11 15:52 source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ rm backup/main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ rmdir backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
docs source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ |

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise

```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir student_portal
angela_jose@LAPTOP-B3CJ092Q:~$ cd student_portal
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ mkdir src docs backup logs config
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls
backup config docs logs src
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tree
.
├── backup
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

6 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ nano logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ head -5 logs/system.og
head: cannot open 'logs/system.og' for reading: No such file or directory
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ head -5 logs/system.log
System Started
User Login
Configuration Loaded
Database Connected
Service Started
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tail -5 logs/system.log
Permissions Verified
File Synced
System Healthy
System Shutdown
Memory Checked

```

```

angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ nano logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tail -5 logs/system.log
Permissions Verified
File Synced
System Healthy
System Shutdown
Memory Checked
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ cp docs/README.txt backup/
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ mv backup/README.txt backup/README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ chmod 600 config/config.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls -l config
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 11 16:05 config.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ chmod 644 docs/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls -l docs
total 0
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 11 16:05 README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ rm backup/README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ rmdir backup
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls
config docs logs src
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ |

```